

Notes: Berry, Levinsohn, and Pakes (JPE, 2004)

Differentiated Products Demand Systems from a Combination of Micro and Macro Data: The New Car Market

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1 Big picture

Question. How can researchers estimate a differentiated-products demand system that gives realistic substitution patterns when the market has many products and consumers differ in their tastes?

Setting. The paper studies the 1993 U.S. new vehicle market. Products are individual car, van, SUV, and pickup models. The key micro data are a proprietary General Motors survey of recent vehicle purchasers, called CAMIP, that reports each buyer's first choice, household attributes, and the buyer's stated second choice.

Core challenge. Aggregate market shares discipline mean utilities, but by themselves they have limited information about which products substitute for which other products. Standard logit models imply substitution patterns that depend mainly on market shares, not product similarity. Observed demographics help, but the paper shows that observed demographics are not enough to reproduce the tight substitution patterns seen in the second-choice data.

Main contribution. Berry, Levinsohn, and Pakes combine macro data on product-level shares with micro data on consumer attributes and second choices. The second-choice data are used as direct evidence about substitution patterns. This lets the authors estimate random coefficients that capture unobserved consumer heterogeneity in tastes for product characteristics.

Main empirical design. The paper estimates a random-coefficients discrete-choice model. The estimation matches three kinds of moments:

- aggregate market shares for each vehicle model;
- covariances between observed household attributes and first-choice vehicle characteristics;
- covariances between first-choice vehicle characteristics and second-choice vehicle characteristics.

Main baseline result. Unobserved random coefficients are essential. Models without unobserved heterogeneity, even when they include observed demographics, produce unreasonable substitution patterns. In contrast, the full model predicts that consumers substitute toward vehicles of similar type, price, size, and brand family.

Important limitation. Second-choice data identify the *pattern* of substitution very well, but they do not by themselves pin down the *absolute level* of price elasticities in a single cross section. The level of price responsiveness still requires instruments, supply-side pricing assumptions, cross-market variation, or outside information such as an assumed aggregate market elasticity.

Policy use. The estimated demand system is used for out-of-sample exercises: introducing high-end SUVs into the 1993 market and removing Oldsmobile from General Motors' product portfolio.

Economic takeaway. Rich micro data can make differentiated-products demand estimation much more credible because they directly discipline substitution. However, realistic substitution patterns and credible price elasticities are distinct objects: the former can be learned from second-choice data, while the latter still requires information about prices, costs, or exogenous variation.

2 Data and measurement: key objects

2.1 Market and products

The empirical market is the U.S. new vehicle market in the 1993 model year. The authors define a product-level choice set of 203 vehicles: 147 cars, 25 SUVs, 17 vans, and 14 pickup trucks.

Interpretation. The large choice set makes the application demanding. Consumers are choosing among many differentiated alternatives, so the central empirical question is not only total demand, but also which particular products are close substitutes.

2.2 Aggregate product-level data

The aggregate data contain product-level market shares and vehicle characteristics. Characteristics include price, horsepower, passenger capacity, fuel economy, safety, accessories, all-wheel drive, vehicle type dummies, payload measures, and firm dummies.

The paper reports about 10.6 million vehicles sold in 1993, at an average transaction price of roughly \$18,500. Total sales were about \$196 billion, and the light-truck market alone accounted for about \$81.2 billion.

Interpretation. Aggregate shares are used to recover mean utility levels for each vehicle. These mean utilities summarize the average attractiveness of each product, including observed characteristics and unobserved quality.

2.3 CAMIP micro data

The main micro data are from CAMIP, a proprietary survey conducted for General Motors. The survey samples vehicle registrations from the 1993 model year and records recent purchasers' choices and household characteristics.

Sample. The subsample used in the paper contains 37,500 observations.

Observed household attributes. CAMIP includes income, age of the household head, family size, number of children, number of adults, and residence type such as urban, suburban, or rural.

Interpretation. These data make it possible to ask whether households with different observable characteristics systematically choose different vehicle characteristics. For example, families with children may value minivans, and rural households may value pickup payload and four-wheel-drive capability.

2.4 Current Population Survey comparison

The authors match CAMIP household variables to similar variables in the Current Population Survey (CPS). The CPS is used to approximate the distribution of household attributes in the population.

Descriptive fact. CAMIP households are much richer than the overall CPS population because new vehicle purchasers are positively selected on income. CAMIP households are also less urban and more rural than the full U.S. household population.

Interpretation. This comparison matters because the model integrates household-level choice probabilities over a population distribution of attributes. CAMIP is choice-based, while CPS helps describe the population that could purchase.

2.5 Second-choice data

The most distinctive data object is the stated second choice: the vehicle a buyer says they would have purchased if their first choice had not been available.

Examples. Accord buyers often list Camry as the modal second choice; Caravan buyers often list Voyager; Grand Cherokee buyers often list Explorer; GMC full-size pickup buyers often list Chevy full-size pickup.

Interpretation. Second choices are direct evidence about substitution. They reveal whether a model predicts substitution toward similar products or toward implausible products simply because those products have high market shares.

2.6 Main dependent objects

The paper does not estimate a simple single-equation demand regression. Instead, it estimates model parameters by matching several moments.

Market-share moments. For every vehicle j , predicted aggregate share should equal observed aggregate share.

First-choice demographic moments. The model should match the covariance between household attributes and the characteristics of the vehicle chosen first.

Second-choice substitution moments. The model should match the covariance between the characteristics of first-choice and second-choice vehicles.

Interpretation. The first set of moments identifies mean utilities. The second set identifies how observed demographics interact with product characteristics. The third set identifies unobserved heterogeneity and the shape of substitution patterns.

2.7 Descriptive facts

Several descriptive facts motivate the model:

- New vehicle buyers are disproportionately high income relative to the population.
- Household attributes are correlated with chosen vehicle characteristics, such as kids with mini-vans and rural residence with pickups.
- First and second choices are strongly correlated in vehicle characteristics: similar products are close substitutes.
- Second choices are often within the same manufacturer, which matters for multiproduct pricing and product-line decisions.

3 Models and empirical specifications: all equations plus interpretation

3.1 Random-coefficients utility

Consumer i 's indirect utility from product j is

$$u_{ij} = \sum_k x_{jk} \tilde{\beta}_{ik} + \xi_j + \varepsilon_{ij}. \quad (1)$$

Objects.

- x_{jk} is observed characteristic k of vehicle j .
- $\tilde{\beta}_{ik}$ is consumer i 's taste for characteristic k .
- ξ_j is unobserved vehicle quality.
- ε_{ij} is an idiosyncratic preference shock.

Interpretation. Consumers value vehicles as bundles of characteristics. The key feature is that tastes are heterogeneous: different consumers can value price, passenger capacity, horsepower, safety, or vehicle type differently.

3.2 Observed and unobserved consumer heterogeneity

The taste coefficient for characteristic k is

$$\tilde{\beta}_{ik} = \bar{\beta}_k + \sum_r z_{ir} \beta_{kr}^o + \beta_k^u \nu_{ik}. \quad (2)$$

Objects.

- $\bar{\beta}_k$ is the average taste for characteristic k .
- z_{ir} is observed household attribute r for consumer i .
- β_{kr}^o measures how observed household attribute r changes taste for characteristic k .
- ν_{ik} is unobserved consumer heterogeneity.
- β_k^u is the standard deviation of unobserved taste for characteristic k .

Interpretation. Observed demographics explain some preference variation, but not all. The β_k^u terms allow consumers with the same observed demographics to differ in their tastes for vehicle characteristics.

3.3 Mean utility and the estimable utility equation

Substituting the taste equation into utility gives

$$u_{ij} = \delta_j + \sum_{k,r} x_{jk} z_{ir} \beta_{kr}^o + \sum_k x_{jk} \nu_{ik} \beta_k^u + \varepsilon_{ij}, \quad (3)$$

where

$$\delta_j = \sum_k x_{jk} \bar{\beta}_k + \xi_j. \quad (4)$$

Interpretation. δ_j is the product-specific mean utility. It absorbs the average value of observed characteristics and the unobserved quality of product j .

Why this matters. The micro data can estimate product-specific constants δ_j together with the heterogeneity parameters β^o and β^u . But decomposing δ_j into the effect of price and other characteristics plus unobserved quality requires additional identifying assumptions.

3.4 Choice probabilities

Conditional on observed demographics z_i and unobserved tastes ν_i , the extreme-value assumption for ε_{ij} implies logit choice probabilities:

$$\Pr(y_i^1 = j \mid z_i, \nu_i, v, x) = \frac{\exp\left(\delta_j + \sum_{k,r} x_{jk} z_{ir} \beta_{kr}^o + \sum_k x_{jk} \nu_{ik} \beta_k^u\right)}{1 + \sum_q \exp\left(\delta_q + \sum_{k,r} x_{qk} z_{ir} \beta_{kr}^o + \sum_k x_{qk} \nu_{ik} \beta_k^u\right)}. \quad (5)$$

Interpretation. Conditional on heterogeneous tastes, choice probabilities have a logit form. Aggregating over heterogeneous consumers generates rich substitution patterns that are not constrained by the plain logit independence-of-irrelevant-alternatives property.

3.5 Aggregate market shares

Let $w = (z, \nu, \varepsilon)$ collect consumer attributes and idiosyncratic shocks. The aggregate share of product j is

$$s_j(\delta, \beta^o, \beta^u; x, P_w) = \int_{A_j(\delta, \beta^o, \beta^u; x)} P_w(dw), \quad (6)$$

where

$$A_j(\delta, \beta^o, \beta^u; x) = \{w : u_{ij}(w) \geq u_{iq}(w) \text{ for all } q = 0, 1, \dots, J\}. \quad (7)$$

Interpretation. Aggregate demand is the fraction of consumers whose tastes and shocks make product j the best choice.

3.6 The BLP contraction and concentration of mean utilities

For any candidate heterogeneity parameters $\beta = (\beta^o, \beta^u)$, the authors choose δ so that predicted shares match observed product shares:

$$s_j^N = s_j(\delta(\beta), \beta; x, P_w), \quad j = 1, \dots, J. \quad (8)$$

Interpretation. This is the BLP contraction idea. Instead of searching directly over hundreds of product-specific constants, the algorithm solves for the vector $\delta(\beta)$ that exactly rationalizes observed market shares for each candidate value of the nonlinear parameters.

Why this matters. The automobile choice set has more than 200 products. Concentrating out δ makes estimation computationally feasible.

3.7 First-choice household-characteristic moments

The model is required to match moments of the form

$$E[x_k^1 z_r], \quad (9)$$

where x_k^1 is characteristic k of the household's first-choice vehicle and z_r is household attribute r .

In sample terms, the paper compares observed moments for buyers of vehicle j to model-predicted conditional moments such as

$$E[z \mid y^1 = j, \beta] = \frac{\int_z \int_\nu z \Pr(y^1 = j \mid z, \nu, \beta) P(dz, d\nu)}{\Pr(y^1 = j \mid \beta)}. \quad (10)$$

Interpretation. These moments identify the observed interactions β^o . For example, they tell the model whether households with children are more likely to buy minivans and whether rural households are more likely to buy pickups.

3.8 Second-choice moments

The second-choice moments compare characteristics of first-choice and second-choice vehicles. A simplified version is

$$E[x_k^1 x_k^2], \quad (11)$$

where x_k^2 is characteristic k of the second-choice vehicle.

For a household whose first choice is j , the model computes the probability that the second choice is q by removing product j and the outside option from the relevant choice set and applying the same logit formula over the remaining products.

Interpretation. If buyers of a minivan usually list another minivan as their second choice, the model must generate enough heterogeneity in tastes for minivans. If buyers of a full-size pickup usually list another full-size pickup, the model must make pickup buyers strongly attached to pickup characteristics.

3.9 Price coefficient and effective wealth

The price coefficient is constrained to be negative by modeling it as

$$\alpha_i = -\exp(-W_i), \quad (12)$$

with

$$W_i = \sum_r z_{ir} \beta_{w,r}^o + \beta_w^u \nu_{iw}. \quad (13)$$

Interpretation. Consumers differ in price sensitivity. Higher effective wealth reduces the absolute disutility of price, while larger families can have lower effective wealth for a vehicle purchase.

3.10 Estimating the average characteristic effects

After estimating δ , β^o , and β^u , the remaining average characteristic effects satisfy

$$\delta_j = p_j \bar{\beta}_p + \sum_{k \neq p} x_{jk} \bar{\beta}_k + \xi_j. \quad (14)$$

Endogeneity problem. Price is likely correlated with unobserved quality ξ_j . Vehicles with high unobserved quality tend to have higher prices. Therefore, regressing δ_j on price without instruments would understate price sensitivity.

Interpretation. Second-choice data can pin down which products are close substitutes, but they do not by themselves fully identify the average price slope in one cross section.

3.11 Pricing equation and alternative elasticity normalizations

One way to help identify $\bar{\beta}_p$ is to use a supply-side pricing equation. Marginal cost is written as

$$mc_j = \sum_k x_{jk} \gamma_k + \omega_j, \quad (15)$$

so equilibrium pricing implies

$$p_j = \sum_k x_{jk} \gamma_k + \text{markup}_j(x, p, \delta, \bar{\beta}_p, \beta^o, \beta^u) + \omega_j. \quad (16)$$

Interpretation. Under a Nash pricing assumption, prices equal marginal costs plus markups. The markup depends on demand elasticities, so the pricing equation provides additional information about the average price coefficient.

The paper compares three choices for $\bar{\beta}_p$:

- $\bar{\beta}_p = 0$, corresponding to ignoring the correlation between product constants and price;
- $\bar{\beta}_p = -3.58$, obtained from a demand-and-pricing-equation approach;
- $\bar{\beta}_p = -11$, chosen to make the aggregate market price elasticity equal to approximately -1 .

Economic message. The level of elasticities changes with $\bar{\beta}_p$, but the pattern of substitution across products is fairly stable.

3.12 Prediction exercises

The paper then uses the estimated demand system for two counterfactual exercises.

Introducing high-end SUVs. The authors introduce hypothetical Mercedes and Toyota SUVs into the 1993 market. The simulated vehicles have SUV characteristics similar to the Ford Explorer but higher quality and price.

Removing Oldsmobile. The authors remove all Oldsmobile products from the choice set and ask where Oldsmobile buyers would go.

Interpretation. These exercises use the estimated substitution structure to predict demand reallocation after product entry or exit. They do not fully solve for all firms' price and product-characteristic responses.

4 Identification and instruments: what makes the estimates credible

4.1 What the micro data identify

The micro data identify how household attributes and unobserved tastes affect vehicle choices, conditional on product-specific constants. In other words, the survey data can estimate β^o , β^u , and δ without first deciding exactly how unobserved vehicle quality ξ_j relates to price and other characteristics.

Credibility point. This is useful because assumptions about the joint distribution of product characteristics and unobserved quality are hard to defend. Product characteristics are chosen by firms, so they are not plausibly random.

4.2 Why second choices identify substitution patterns

Second-choice data are especially powerful because they directly report where consumers say they would substitute if their chosen product disappeared. This is much closer to the economic object of interest than aggregate shares alone.

Example. If Grand Cherokee buyers usually choose Explorer as their second choice, the demand model must make those two vehicles close substitutes. If GMC full-size pickup buyers often choose Chevy full-size pickup, the model must capture within-truck and within-firm substitution.

Economic logic. Random coefficients create groups of consumers who have strong tastes for particular characteristics. These consumers substitute toward products that share those characteristics.

4.3 Why observed demographics are not enough

Observed household attributes explain some first-choice patterns. For example, children help explain minivan choices, and rural residence helps explain pickup choices. But observed demographics do not explain all the persistent taste differences that generate tight second-choice substitution.

Evidence. The logit models with observed demographics fit the first-choice demographic moments fairly well but do poorly in predicting second-choice substitution. They often predict implausible substitutes such as a minivan or full-size pickup for many ordinary passenger cars.

Interpretation. Large data sets with demographics are valuable, but substitution depends on latent needs and tastes: commute length, taste for brands, need for cargo space, style preferences, and other factors not recorded in the survey.

4.4 Why price elasticities need additional information

Second-choice data identify relative substitution conditional on changing the choice set. But a small price increase does not force all consumers to leave the product; only relatively price-sensitive consumers leave. Therefore, the absolute level of price elasticity depends on the average price coefficient and its distribution.

Key limitation. In a single cross section, the model cannot estimate the average price coefficient precisely from second choices alone. There is no repeated observation of the same product at different prices.

Solution in the paper. The authors use additional assumptions: instrumental variables for the demand equation, a supply-side pricing equation, and a calibration based on an aggregate market elasticity suggested by industry knowledge.

4.5 Price endogeneity and instruments

The demand-side equation for mean utility contains unobserved product quality:

$$\xi_j = \delta_j - p_j \bar{\beta}_p - \sum_{k \neq p} x_{jk} \bar{\beta}_k. \quad (17)$$

The core moment restriction is that instruments based on observed characteristics are orthogonal to unobserved quality:

$$E[Z_j \xi_j] = 0. \quad (18)$$

Interpretation. Instruments are needed because price is not randomly assigned. High-quality vehicles can command high prices, and failing to account for this makes demand look too inelastic.

4.6 Comparison models as specification tests

The paper uses comparison models to show what goes wrong when key ingredients are removed.

Logit with first-choice data. Allows observed demographics but no unobserved random coefficients. It predicts poor substitution patterns.

Logit with first- and second-choice data. Still fails because it lacks unobserved heterogeneity.

Model with no observed attributes. Keeps random coefficients but removes observed demographic interactions. It does surprisingly well for substitution patterns, showing that unobserved heterogeneity is more important than observed demographics for substitution.

Full model. Includes both observed and unobserved heterogeneity and performs well on both first-choice demographic moments and second-choice substitution moments.

4.7 Limitations

Single-year market. The main data are from the 1993 market, so the paper cannot rely heavily on time variation in product sets and prices.

Second-choice survey limitations. Consumers report one second choice, not a complete ranking. They also are not asked whether they would leave the new vehicle market entirely if the first choice were unavailable.

Counterfactual limitations. In the product-entry and product-exit exercises, the paper does not fully model all rival price responses or future product redesign decisions.

Interpretation. The paper is strongest as evidence that second-choice data identify realistic substitution patterns. It is more cautious about the absolute level of price elasticities and complete equilibrium counterfactuals.

5 Tables and figures

5.1 Table 1: comparison of consumer samples

TABLE 1
COMPARISON OF CONSUMER SAMPLES

Income Range	Percentage in CPS	Percentage in CAMIP	CPS Group Mean	CAMIP Mean
0–36,500	64.17	25.00	16.90	25.96
36,500–55,000	16.97	23.16	44.89	45.43
55,000–85,000	12.34	26.71	66.93	67.46
85,000–	6.52	25.13	114.25	148.19
All	100.00	100.00	34.17	72.27
Other Demographics				
Family size			2.36	2.65
Age of household head			46.80	46.18
Number of kids			.66	.58
Urban			.46	.35
Rural			.25	.35
Suburban			.29	.30

Figure 1: Table 1: CPS and CAMIP consumer samples.

Read:

- CAMIP purchasers are much higher income than the general CPS population.
- The mean income in CAMIP is about \$72,270, compared with about \$34,170 in the CPS.
- CAMIP households are less urban and more rural than the overall household population.
- Family size and age of household head are similar across the two samples.

Economic message. New vehicle buyers are selected. Demand estimation must account for the distribution of buyers and the population distribution of household attributes, rather than treating the buyer survey as a random sample of all households.

5.2 Table 5: examples of second choices

TABLE 5
EXAMPLES OF SECOND CHOICES

Model	n_j	Modal Second Choice	Number Choosing	Next Second Choice	(Modal+Next)/ n	Number of Different Choices
Metro	188	Escort	22	Geo Storm	.22	49
Cavalier	238	Escort	16	LeBaron	.12	59
Escort	166	Tempo	16	Taurus	.18	53
Corolla	250	Civic	42	Camry	.33	55
Sentra	203	Corolla	34	Civic	.31	60
Accord	223	Camry	58	Taurus	.35	61
Taurus	147	Camry	18	Sable	.22	45
Legend	119	Lexus ES300	19	Lexus SC300	.24	40
Seville	243	DeVille	38	Lincoln MK8	.26	49
Lexus LS400	148	DeVille	33	Infiniti Q45	.39	27
Caravan	166	Voyager	31	Aerostar	.32	36
Quest	232	Caravan	50	Villager	.43	31
Grand Cherokee	137	Explorer	75	Blazer	.59	34
Trooper	137	Explorer	43	Rodeo	.41	27
GMC Full-Size Pickup	469	Chevy Full-Size Pickup	222	Ford Full-Size Pickup	.55	29
Toyota Pickup	113	Ford Ranger	29	Nissan Pickup	.43	25
Econovan	90	Chevy Full-Size Van	20	Suburban	.44	23

Figure 2: Table 5: second-choice vehicles for selected first choices.

Read:

- Second choices are often very intuitive: Accord to Camry, Grand Cherokee to Explorer, Caravan to Voyager, GMC full-size pickup to Chevy full-size pickup.
- The concentration of second choices is especially high for trucks and SUVs.
- The modal and next-best second choices often account for a large share of responses.
- Many second choices are within the same firm, which matters for product-line pricing.

Economic message. Second-choice data reveal close substitutes directly. They provide the empirical discipline that aggregate market shares alone usually cannot provide.

5.3 Tables 6 and 7: observed and unobserved heterogeneity

TABLE 6
ESTIMATES OF INTERACTION TERMS, β^o

VEHICLE CHARACTERISTIC AND HOUSEHOLD ATTRIBUTE	FULL MODEL (1)	LOGIT	
		First (2)	First and Second (3)
Price:			
Constant	-2.18 (.142)	.092 (.0001)	.139 (.0003)
Income $\times$ (income <75th percentile)	.714 (.044)	.299 (.002)	.344 (.001)
Income $\times$ (income >75th percentile)	1.17 (.083)	.466 (.091)	.603 (.007)
Family size	-.565 (.010)	-.144 (.001)	-.143 (.006)
Minivan: Kids (kids have age ≤ 16)	1.973 (.242)	.765 (.098)	.771 (.323)
Pass:			
Adults (adults have age >16)	.203 (.095)	.018 (.0004)	-.067 (.009)
Family size	.536 (.052)	-.055 (.003)	-.006 (.0002)
Age (of household head)	.019 (.003)	.002 (.00001)	.005 (.00001)
HP: Age	-.002 (.001)	-.010 (.0004)	-.012 (.0001)
Acc:			
Age	.0004 (.001)	.001 (.00001)	-.002 (.0001)
Age ²	.0001 (.00001)	.000 (.00001)	.000 (.00001)
PUPayl:			
Age	.0174 (.002)	-.003 (.0001)	.000 (.00001)
Rural dummy	1.075 (.179)	.512 (.005)	.376 (.008)
Safe: Age	.013 (.0006)	.015 (.001)	.016 (.0004)
SUV:			
Age	-.219 (.010)	-.043 (.003)	-.043 (.004)
Rural dummy	.332 (.156)	.403 (.007)	-.016 (.002)
Allw: Rural dummy	.278 (.247)	.142 (.005)	.734 (.246)
Outside good:			
Total income	5.151 (.228)	-.228 (.096)	-.305 (.063)
Family size	-.007 (.002)	.532 (.057)	-.346 (.004)
Adults	-.428 (.766)	.851 (.112)	1.953 (.148)

(a) Observed interactions β^o .

TABLE 7
ESTIMATES OF INTERACTION TERMS, β^u

Parameter Name	Full Model (1)	$\beta^u = 0$ (2)
Price	.449 (.026)	.055 (.004)
HP	.030 (.016)	.183 (.020)
Pass	2.74 (.147)	1.444 (.055)
Sport	.002 (.0004)	2.763 (.068)
Acc	.554 (.078)	.515 (.055)
Safe	.260 (.130)	.376 (.093)
MPG	.488 (.018)	.430 (.017)
Allw	.740 (.179)	.431 (.049)
Minivan	4.787 (.353)	6.641 (.113)
SUV	3.076 (.292)	3.231 (.114)
Van	1.713 (.289)	6.888 (.266)
PUPayl	2.160 (.092)	4.301 (.210)
SUVPayl	.356 (.072)	.015 (.013)
Chrysler	1.689 (.058)	1.383 (.051)
Ford	.915 (.072)	1.410 (.051)
GM	1.885 (.057)	1.844 (.105)
Honda	.329 (.128)	.086 (.043)
Nissan	.506 (.142)	1.588 (.071)
Toyota	.169 (.134)	.576 (.094)
Small Asian*	1.467 (.068)	2.155 (.022)
European*	.454 (.084)	1.883 (.034)
Outside good	27.858 (1.004)	10.256 (.506)

* We constrained the coefficients on the dummies for the different European firms to be the same, and we did the same for the smaller Asian producers.

(b) Unobserved interactions β^u .

Figure 3: Tables 6 and 7: parameter estimates for observed and unobserved taste heterogeneity.

Read:

- Table 6 shows sensible observed interactions: kids raise taste for minivans, rural households value pickup payload, and older households value safety more.
- The price coefficient varies with income and family size through the effective-wealth specification.
- Table 7 shows large and significant unobserved heterogeneity for many characteristics, including passenger capacity, minivans, SUVs, vans, pickups, and firm dummies.
- The comparison with the $\beta^o = 0$ model shows that unobserved heterogeneity alone can explain much of the substitution structure.

Economic message. Observed demographics matter for who buys what, but unobserved heterogeneity is central for substitution patterns. Consumers differ in many ways not captured by standard demographic variables.

5.4 Table 8: level of price elasticities

TABLE 8 IMPLICATIONS OF ALTERNATIVE ESTIMATES OF β_p			
	VALUE OF $\bar{\beta}_p$		
	0	-3.58	-11
A. Implied Average across Vehicles			
Mean semi-elasticity	-.75	-3.94	-10.56
Total market elasticity	-.2	-.4	-1
B. Coefficients from Projecting Semi-elasticities			
Price	-.016 (.003)	-.031 (.006)	-.063 (.014)
HP	.023 (.025)	-.025 (.044)	-.122 (.102)
Pass	.023 (.029)	.057 (.052)	.127 (.121)
Sport	-.235 (.069)	-.230 (.117)	-.219 (.273)
Acc	-.086 (.023)	-.066 (.040)	-.023 (.093)
Safe	-.177 (.038)	-.137 (.067)	-.052 (.126)
MPG	.010 (.007)	-.034 (.013)	-.126 (.029)
Allw	.084 (.103)	.275 (.182)	.671 (.425)
Minivan	-.174 (.099)	-.730 (.174)	-1.882 (.406)
SUV	-.480 (.179)	-.923 (.316)	-1.841 (.735)
Van	-.339 (.154)	-1.112 (.272)	-2.714 (.633)
PUPayl	-.173 (.050)	-.625 (.088)	-1.562 (.204)
SUVPayl	-.107 (.101)	-.058 (.144)	-.400 (.416)

NOTE.—Firm dummies are suppressed.

Figure 4: Table 8: implications of alternative values of $\bar{\beta}_p$.

Read:

- The paper compares $\bar{\beta}_p = 0$, $\bar{\beta}_p = -3.58$, and $\bar{\beta}_p = -11$.
- The mean semi-elasticity becomes much larger in absolute value as $\bar{\beta}_p$ becomes more negative.
- The implied aggregate market elasticity ranges from about -0.2 to -1 .
- Vehicle-type patterns remain sensible: trucks, vans, and SUVs tend to have lower elasticities,

consistent with higher markups.

Economic message. Second-choice data discipline substitution patterns, but the overall price-elasticity level still depends on additional identifying information.

5.5 Table 9: price substitutes in the full model

TABLE 9
PRICE SUBSTITUTES FOR SELECTED VEHICLES: ESTIMATES FROM THE FULL MODEL

Vehicle	Price	Semi-elasticity	Best Substitute	Price	Movers* (%)	Second Best	Price	Movers* (%)	To Outside† (%)
Metro	7.84	-1.77	Tercel	9.70	14.96	Festiva	7.41	10.57	17.96
Cavalier	11.46	-4.08	Escort	11.49	8.62	Tempo	10.78	6.80	6.81
Escort	11.49	-4.02	Tempo	10.78	8.21	Cavalier	11.49	7.29	6.56
Corolla	14.51	-3.92	Civic	14.00	8.08	Escort	11.49	7.91	5.00
Sentra	11.78	-3.79	Civic	14.00	13.36	Escort	11.49	4.70	6.55
Accord	17.25	-3.92	Camry	18.20	8.60	Civic	13.00	4.47	5.06
Taurus	17.65	-3.73	Accord	17.25	6.25	Mercury Sable	18.66	6.09	3.97
Legend	32.42	-3.73	Accord	17.25	3.96	Camry	18.20	3.87	4.38
Seville	43.83	-3.16	DeVille	34.40	10.12	El Dorado	35.74	8.04	5.57
Lexus LS400	51.29	-3.43	Mercedes 300	47.71	7.97	Lincoln Town Car	35.68	6.29	5.87
Caravan	17.56	-3.32	Voyager	17.59	35.11	Aerostar	18.13	10.19	5.20
Quest	20.55	-3.98	Aerostar	18.13	12.50	Caravan	17.56	10.38	5.48
Grand Cherokee	25.84	-3.06	Explorer	24.27	17.60	Cherokee	20.10	9.51	6.38
Trooper	22.78	-3.96	Explorer	24.27	17.53	Grand Cherokee	25.85	8.50	5.42
GMC Full-Size Pickup	16.76	-3.78	Chevy Full-Size Pickup	16.78	43.74	Ford Full-Size Pickup	16.68	13.56	6.03
Toyota Pickup	13.77	-3.34	Ranger	11.74	20.53	Nissan Pickup	11.10	11.93	9.35
Econovan	24.54	-2.86	Chevy Van	25.96	12.90	Dodge Van	23.71	9.73	5.38

* Of those who substitute away from the given good in response to the price change, the fraction who substitute to this good.
† Of those who substitute away from the given good in response to the price change, the fraction who substitute to the outside good.

Figure 5: Table 9: price substitutes predicted by the full model.

Read:

- When a vehicle's price rises, the best substitutes are usually vehicles of the same type and similar price.
- Corolla substitutes toward Civic and Escort; Caravan substitutes strongly toward Voyager; Grand Cherokee substitutes toward Explorer.
- Full-size pickups substitute strongly toward other full-size pickups.
- The outside-good share of switchers is generally modest in these local price-change experiments.

Economic message. The full model generates substitution patterns that look like actual consumer substitution, not mechanical market-share reallocation.

5.6 Tables 10 and 11: model comparison and fit to second choices

TABLE 10
PRICE SUBSTITUTES FOR SELECTED VEHICLES: A COMPARISON AMONG MODELS

VEHICLE	FULL MODEL	LOGIT		SIGMA ONLY
		First	First and Second	
Metro	Tercel	Caravan	Ford Full-Size Pickup	Civic
Cavalier	Escort	Caravan	Ford Full-Size Pickup	Escort
Escort	Tempo	Caravan	Ford Full-Size Pickup	Ranger
Corolla	Escort	Caravan	Ford Full-Size Pickup	Civic
Sentra	Civic	Caravan	Ford Full-Size Pickup	Civic
Accord	Camry	Caravan	Ford Full-Size Pickup	Camry
Taurus	Accord	Caravan	Ford Full-Size Pickup	Accord
Legend	Town Car	Caravan	Ford Full-Size Pickup	Town Car
Seville	DeVille	Caravan	Ford Full-Size Pickup	DeVille
Lexus LS400	Mercedes 300	Econovan	Ford Full-Size Pickup	Seville
Caravan	Voyager	Voyager	Voyager	Voyager
Quest	Aerostar	Caravan	Caravan	Aerostar
Grand Cherokee	Explorer	Caravan	Chevy Full-Size Pickup	Explorer
Trooper	Explorer	Caravan	Chevy Full-Size Pickup	Rodeo
GMC Full-Size Pickup	Chevy Full-Size Pickup	Caravan	Chevy Full-Size Pickup	Chevy Full-Size Pickup
Toyota Pickup	Ranger	Caravan	Chevy Full-Size Pickup	Ranger
Econovan	Dodge Van	Caravan	Ford Full-Size Pickup	Dodge Van

(a) Price substitute comparison across models.

TABLE 11
MOST POPULAR SECOND CHOICES: A COMPARISON AMONG MODELS AND TO THE DATA

Vehicle	Full Model	Rank	Logit First	Rank	Logit First and Second	Rank	$\beta^* \equiv 0$	Rank
Metro	Chevy Geo Storm	2	Ford Full-Size Pickup	≥ 25	Ford Full-Size Pickup	≥ 25	Tercel	12
Cavalier	Sun Bird	3	Ford Full-Size Pickup	≥ 25	Ford Full-Size Pickup	≥ 25	Ford Escort	1
Escort	Tempo	1	Ford Full-Size Pickup	≥ 25	Ford Full-Size Pickup	≥ 25	Tempo	1
Corolla	Escort	6	Ford Full-Size Pickup	≥ 25	Ford Full-Size Pickup	≥ 25	Civic	1
Sentra	Civic	2	Ford Full-Size Pickup	≥ 25	Ford Full-Size Pickup	≥ 25	Civic	2
Accord	Camry	1	Ford Full-Size Pickup	≥ 25	Ford Full-Size Pickup	≥ 25	Camry	1
Taurus	Mercury Sable	2	Ford Full-Size Pickup	≥ 25	Ford Full-Size Pickup	≥ 25	Accord	4
Legend	Civic	10	Ford Full-Size Pickup	≥ 25	Ford Full-Size Pickup	≥ 25	Town Car	≥ 25
Seville	Deville	1	Ford Full-Size Pickup	≥ 25	Ford Full-Size Pickup	≥ 25	Deville	1
Lexus LS400	MB 300	3	Ford Full-Size Pickup	≥ 25	Ford Full-Size Pickup	≥ 25	DeVille 2	1
Caravan	Voyager	1	Ford Full-Size Pickup	≥ 25	Voyager	1	Voyager	1
Quest	Aerostar	7	Ford Full-Size Pickup	≥ 25	Caravan	1	Caravan	1
Grand Cherokee	Explorer	1	Chevy Full-Size Pickup	≥ 25	Chevy Full-Size Pickup	≥ 25	Explorer	1
Trooper	Explorer	1	Chevy Full-Size Pickup	22	Chevy Full-Size Pickup	22	Rodeo	2
GMC Full-Size Pickup	Chevy Full-Size Pickup	1	Chevy Full-Size Pickup	1	Ford Full-Size Pickup	2	Chevy Full-Size Pickup	1
Toyota Pickup	Ranger	1	Chevy Full-Size Pickup	4	Chevy Full-Size Pickup	4	Ranger	1
Econovan	Chevy Van	1	Ford Full-Size Pickup	6	Ford Full-Size Pickup	6	Chevy Van	1

(b) Predicted most popular second choices versus the data.

Figure 6: Tables 10 and 11: comparison of substitution patterns across specifications.

Read:

- Logit models often predict implausible best substitutes, such as Caravan or Ford full-size pickup for many passenger cars.
- The full model predicts sensible substitutes such as Accord to Camry, Seville to DeVille, and Grand Cherokee to Explorer.
- In Table 11, the full model matches the actual modal second choice in 9 of 17 cases.
- The full model predicts one of the top three second choices in 15 of 17 cases and never predicts a best substitute ranked worse than tenth in the data.

Economic message. These tables are the strongest evidence that random coefficients and second-choice moments matter. The main empirical success of the paper is realistic substitution.

5.7 Tables 12 and 13: counterfactual predictions

TABLE 12
INTRODUCING A MERCEDES SUV

Model	Price	Old Share	New Share	New – Old Share
New car	33,659	.0000	.0762	.0762
Biggest Declines in Sales				
Ford Explorer	24,274	.2518	.2373	–.0144
Jeep Grand Cherokee	25,849	.1475	.1376	–.010
Chevy S10 Blazer	22,651	.1106	.1071	–.0036
Toyota 4Runner	25,548	.0380	.0347	–.0033
Nissan Pathfinder	24,943	.0397	.0375	–.0022
Luxury cars*		.1610	.1565	–.0045
All vehicles		9.711	9.711	.000

NOTE.—Characteristics of the new car are described in the text.
* Cars priced above \$30,000.

(a) Introducing a Mercedes SUV.

TABLE 13
DISCONTINUING THE OLDSMOBILE DIVISION

	Old Share	New Share	New – Old Share
All Oldsmobiles	.237	0	–.237
All GM	3.126	3.016	–.110
All cars	9.711	9.695	–.016
Non-Olds Share Changes			
Chevy Lumina	.1354	.1548	.0194
Buick LeSabre	.1216	.1336	.0120
Pontiac GrandAm	.1322	.1441	.0119
Honda Accord	.2955	.3039	.0084
Ford Taurus	.2040	.2115	.0075
Saturn SL	.1465	.1539	.0074
Toyota Camry	.2343	.2415	.0072
Buick Century	.0614	.0683	.0069
Pontiac Grand Prix	.0517	.0584	.0067
Chevy Cavalier	.1700	.1767	.0067
Pontiac Bonneville	.0658	.0721	.0064

NOTE.—The original Oldsmobile models in the data (and their shares) are Circa (.068), Cutlass Supreme (.059), Olds 88 (.050), Achieva (.033), Olds 98 (.019), and Bravada (.008).

(b) Discontinuing Oldsmobile.

Figure 7: Tables 12 and 13: product-entry and product-exit exercises.

Read:

- The hypothetical Mercedes SUV captures a predicted share of 0.0762, drawing mostly from other SUVs and some luxury cars.
- The largest losses are for Explorer, Grand Cherokee, Chevy S10 Blazer, Toyota 4Runner, and Nissan Pathfinder.
- Removing Oldsmobile reduces all-GM share from 3.126 to 3.016 in the units used in the table.
- The biggest non-Oldsmobile gainers are GM family cars such as Chevy Lumina, Buick LeSabre, and Pontiac GrandAm, but some buyers switch to Honda Accord, Ford Taurus, Toyota Camry, and other non-GM models.

Economic message. The value of a demand system is not only estimating elasticities. It also predicts how demand reallocates after product entry, product exit, and product-line restructuring.

6 Short summary

- The paper estimates a random-coefficients differentiated-products demand model for the 1993 U.S. new vehicle market.
- The key innovation is using second-choice survey data to directly discipline substitution patterns.
- Aggregate market shares identify product mean utilities through the BLP contraction.
- First-choice household-characteristic moments identify observed demographic interactions.
- Second-choice moments identify unobserved random coefficients and realistic substitution.
- Logit models with observed demographics fit some demographic choice patterns but fail badly on substitution.

- The full model predicts close substitutes that are similar in type, price, size, and brand/family position.
- Second-choice data do not by themselves identify the absolute level of price elasticities in a single cross section.
- The paper therefore uses additional information from instruments, supply-side pricing restrictions, and elasticity calibration.
- The estimated system is used to predict demand for high-end SUVs and the effect of shutting down Oldsmobile.

7 Conclusion

7.1 Core conclusion: what the paper establishes

Berry, Levinsohn, and Pakes show that micro data on second choices can substantially improve differentiated-products demand estimation. These data provide direct evidence about substitution patterns and make it possible to estimate random coefficients that generate realistic reallocations of demand across products.

The paper also establishes a clear distinction between two identification problems. Second-choice data are powerful for identifying *which* products are close substitutes. They are not sufficient, in a single cross section, to identify the *overall level* of price elasticities.

7.2 Economic intuition

Consumers do not randomly move from one car to another when a product becomes more expensive or disappears. Minivan buyers tend to consider other minivans; SUV buyers tend to consider other SUVs; full-size pickup buyers tend to consider other full-size pickups. These patterns arise because consumers have heterogeneous tastes for product characteristics.

Observed demographics reveal some of this heterogeneity, but much of it is latent. The second-choice data reveal it directly. This is why the random-coefficients model succeeds where simple logit models fail.

7.3 Takeaway

The paper's main lesson is methodological and economic: demand estimation is much stronger when it combines aggregate market shares with micro evidence on consumer substitution. For empirical industrial organization, the paper shows how survey data can discipline counterfactual predictions about entry, exit, and product-line decisions. For applied work more broadly, it shows that the right micro moments can turn a flexible model into a credible economic tool.